



# Information Technology

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# What is Information Technology?

Information Technology (IT) is the branch of technology that focuses on using computers, software, communication systems, and digital networks to create, process, store, and exchange information.

## Key Points:

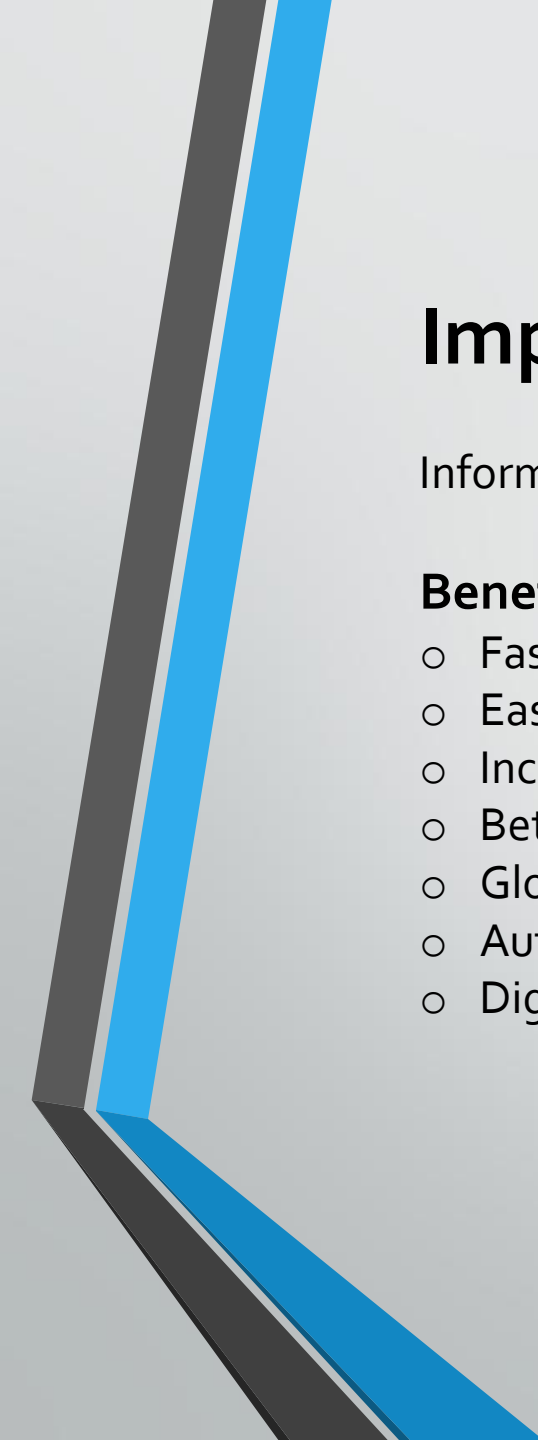
- Uses computers to solve problems
- Supports communication and collaboration
- Helps store and manage data
- Enables internet-based services
- Used in almost every industry

**Example:** Online banking, online shopping, digital classrooms.

# IT vs Computer Science

Although Information Technology (IT) and Computer Science (CS) are closely related, they focus on different aspects of computing. IT emphasizes the practical use, maintenance, and management of technology, while Computer Science focuses on designing and developing software, algorithms, and computing systems.

| Information Technology (IT)                                                                               | Computer Science (CS)                                                             |
|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Focuses on applying technology to solve business and user needs.                                          | Focuses on the theory and principles of computing.                                |
| Manages computer systems, networks, databases, and security.                                              | Develops software, algorithms, and programming solutions.                         |
| Involves system administration, technical support, and IT infrastructure.                                 | Involves software engineering, artificial intelligence, and data structures.      |
| Works with existing technologies and enterprise systems.                                                  | Creates new technologies and innovative software solutions.                       |
| Typical careers: Network Administrator, System Administrator, IT Support Engineer, Cybersecurity Analyst. | Typical careers: Software Developer, Data Scientist, AI Engineer, Game Developer. |

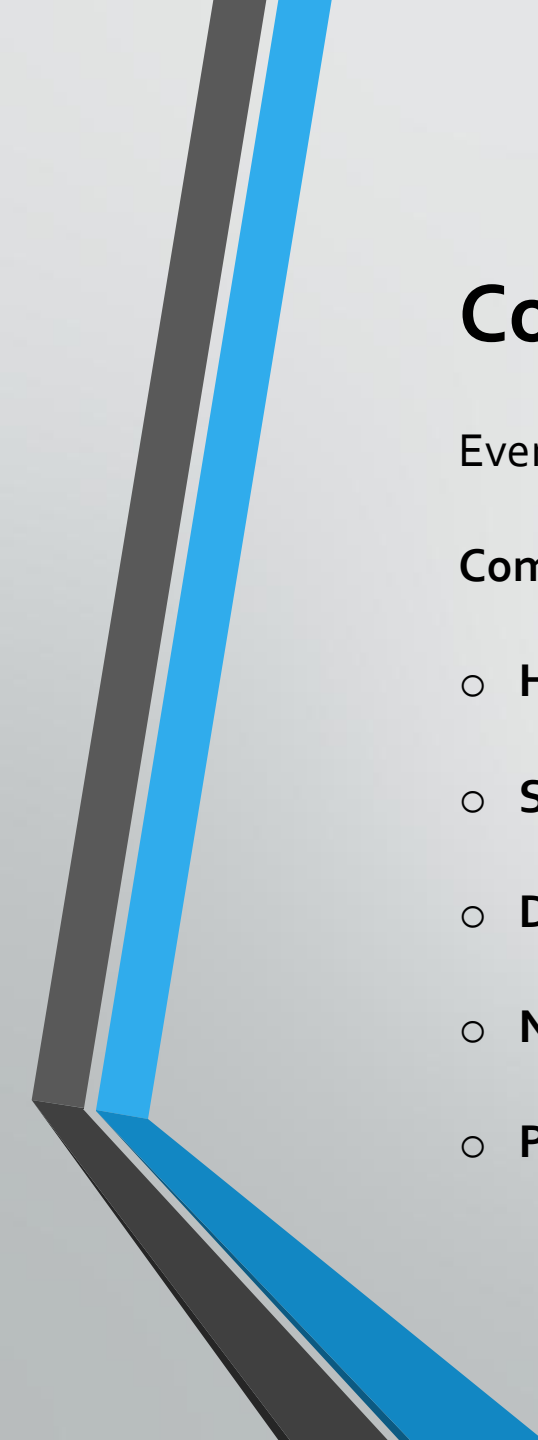


# Importance of Information Technology

Information Technology has transformed the way people work, study, and communicate.

## **Benefits:**

- Faster communication
- Easy access to information
- Increased business productivity
- Better decision making
- Global connectivity
- Automation of repetitive tasks
- Digital services available 24/7



# Core Components of Information Technology

Every IT system consists of five major components.

## Components:

- **Hardware:** Physical devices
- **Software:** Programs and applications
- **Data:** Raw facts processed into information
- **Networks:** Connect computers together
- **People:** Users and IT professionals who operate the system

# Computer Hardware

Hardware refers to all physical parts of a computer that can be touched.

## Examples:

- **Keyboard:** Used to enter text, numbers, and commands into the computer.
- **Mouse:** A pointing device used to select, click, and move items on the screen.
- **Monitor:** Displays text, images, videos, and the computer's output.
- **CPU:** The "brain" of the computer that processes instructions and performs calculations.
- **Motherboard:** The main circuit board that connects all hardware components together.
- **RAM:** Temporarily stores data and programs currently being used by the CPU.
- **Hard Disk:** Stores data permanently, including the operating system, applications, and files.
- **Printer:** Produces a printed (hard) copy of digital documents and images.
- **Scanner:** Converts printed documents or images into digital format.

# Computer Software

Software is a collection of programs that instruct hardware to perform specific tasks.

## Characteristics:

- Cannot be touched
- Controls hardware
- Performs calculations
- Executes user commands

## Types of Software

### System Software

- Operating Systems
- Device Drivers
- Utility Programs

### Application Software

- MS Word
- Excel
- PowerPoint
- Web Browsers
- Media Players

# Computer Networks

## What is a Computer Network?

A computer network is a collection of interconnected devices that communicate with each other to exchange data and share resources. Networks allow users to access files, printers, applications, and internet services from different locations.

## Types of Computer Networks:

Different networks are designed based on geographical coverage.

| Network | Coverage Area      | Example             |
|---------|--------------------|---------------------|
| PAN     | Personal Area      | Bluetooth devices   |
| LAN     | Building or Office | School computer lab |
| MAN     | City               | Cable TV network    |
| WAN     | Country or World   | Internet            |

## Key Point

As the coverage area increases, the complexity and cost of the network also increase.



# Internet and World Wide Web

## Internet

The Internet is a worldwide network of connected computers that enables communication and information sharing.

## World Wide Web (WWW)

The World Wide Web is a collection of websites and web pages that operate over the Internet.

## Network Devices

Network devices help computers communicate efficiently.

### Router

Connects different networks and directs data packets.

### Switch

Connects multiple devices within a LAN.

### Hub

Broadcasts data to all connected devices.

### Modem

Converts digital data into signals for Internet communication.

### Access Point

Provides wireless network connectivity.

# Database Management System (DBMS)

A Database Management System is software used to organize, store, retrieve, and manage large volumes of data efficiently.

## Advantages

- Fast data retrieval
- Data security
- Reduced redundancy
- Easy backup and recovery
- Multi-user access

## Popular DBMS

- MySQL
- Oracle
- Microsoft SQL Server
- PostgreSQL
- MongoDB

# SQL (Structured Query Language)

SQL is the standard language used to interact with relational databases.

## Common SQL Commands:

- **SELECT:** Retrieve records
- **INSERT:** Add new records
- **UPDATE:** Modify existing data
- **DELETE:** Remove records
- **CREATE:** Create database objects

## Applications

- Banking
- Hospital systems
- E-commerce websites
- College Management Systems

# Information Security

Information Security focuses on protecting digital information from unauthorized access, misuse, or destruction.

## Objectives:

**Confidentiality:** Only authorized users can access information.

**Integrity:** Data remains accurate and unchanged.

**Availability:** Information is accessible whenever needed.

## Cybersecurity Threats

Cybersecurity threats are malicious activities that target computer systems and networks.

### Common Threats

- Malware
- Virus
- Worm
- Trojan Horse
- Ransomware
- Phishing
- Spyware
- Denial-of-Service (DoS) Attack

## Cybersecurity Protection Methods

Organizations use multiple security techniques to protect information.

### Security Measures

- Strong Passwords
- Multi-Factor Authentication
- Firewalls
- Antivirus Software
- Data Encryption
- Regular Software Updates
- Secure Backups
- Employee Awareness Training

# Cloud Computing

Cloud Computing allows users to access computing resources through the Internet instead of owning physical servers.

## Characteristics

- On-demand services
- Pay-as-you-use
- Remote accessibility
- High scalability
- Automatic updates

## Cloud Service Models

### Infrastructure as a Service (IaaS)

Provides virtual servers, storage, and networking.

**Example:** Amazon EC2

### Platform as a Service (PaaS)

Provides tools for software development without managing infrastructure.

**Example:** Google App Engine

### Software as a Service (SaaS)

Delivers software through a web browser.

#### Examples

- Gmail
- Microsoft 365
- Google Docs
- Zoom

# Emerging IT Technologies

## ❑ Artificial Intelligence (AI)

Artificial Intelligence enables computers to perform tasks that normally require human intelligence.

### Applications

- Voice Assistants
- Face Recognition
- Medical Diagnosis
- Self-driving Cars
- Chatbots
- Recommendation Systems

## ❑ Internet of Things (IoT)

The Internet of Things connects everyday physical devices to the Internet, allowing them to collect and exchange data.

### Examples

- Smart Watches
- Smart TVs
- Smart Homes
- Smart Traffic Systems
- Smart Agriculture
- Connected Vehicles

# Emerging IT Technologies

## ❑ Blockchain

Blockchain is a decentralized digital ledger that securely records transactions across multiple computers.

### Applications

- Cryptocurrency
- Supply Chain
- Digital Identity
- Healthcare
- Voting Systems

## ❑ Big Data

Big Data refers to extremely large datasets that require advanced tools for storage and analysis.

### Characteristics (5 Vs)

- Volume
- Velocity
- Variety
- Veracity
- Value



# Applications of Information Technology

Information Technology has transformed every sector of society.

## **Business**

- Online Banking
- ERP Systems
- Customer Management

## **Healthcare**

- Electronic Medical Records
- Telemedicine

## **Education**

- E-learning Platforms
- Virtual Classrooms

## **Government**

- Digital Services
- Online Tax Filing

## **Entertainment**

- Streaming Platforms
- Online Gaming

## **Transportation**

- GPS Navigation
- Online Ticket Booking



# Key roles of IT in modern business:

- **Increased efficiency:** IT automates repetitive tasks, reduces manual labor, and minimizes errors, which leads to increased productivity and consistency across departments.
- **Improved communication:** It provides tools like email, instant messaging, and video conferencing that facilitate real-time communication and collaboration among employees, clients, and partners.
- **Data-driven decision-making:** Advanced analytics tools allow businesses to collect, analyze, and interpret vast amounts of data to gain insights for strategic planning and performance monitoring.
- **Innovation and competitiveness:** IT provides platforms and tools for research and development, allowing companies to create innovative products and services and gain a competitive edge.

# Key roles of IT in modern business:

- **Supply chain optimization:** Sophisticated software helps businesses track inventory, manage logistics, and improve demand forecasting, which reduces costs and ensures timely deliveries.
- **Enhanced customer experience:** IT systems like CRM software allow businesses to track customer information and preferences, while tools like chatbots provide instant customer service.
- **Cybersecurity:** IT professionals use advanced security measures like firewalls, encryption, and intrusion detection systems to protect sensitive data from cyber threats and safeguard the company's reputation.
- **Scalability and flexibility:** Cloud computing, for example, provides businesses with scalable storage and access to data from anywhere, allowing them to grow and adapt their operations as needed.

# Enterprise Resource Planning (ERP)

## What is ERP?

Enterprise Resource Planning (ERP) is integrated software that helps organizations manage various business processes using a single centralized system.

## Main Objectives

- Integrate business departments
- Improve efficiency
- Reduce duplicate data
- Automate daily operations
- Support better decision-making

## Popular ERP Software

- SAP ERP
- Oracle ERP
- Microsoft Dynamics 365
- Odoo
- NetSuite

# Core Modules of ERP

ERP consists of different modules, each designed to manage a specific business function.

| Module                           | Purpose                                  |
|----------------------------------|------------------------------------------|
| Finance                          | Manages accounting and financial records |
| Human Resources                  | Employee information and payroll         |
| Inventory                        | Tracks stock and warehouse activities    |
| Sales                            | Customer orders and billing              |
| Procurement                      | Purchasing materials and supplies        |
| Manufacturing                    | Production planning and scheduling       |
| Customer Relationship Management | Customer interactions and support        |

# Benefits of ERP

Implementing ERP offers several advantages to organizations.

## Benefits

- Centralized data management
- Improved productivity
- Better communication
- Real-time reporting
- Lower operational costs
- Accurate financial records
- Better inventory control
- Increased customer satisfaction

## Example

When a customer places an order, ERP automatically updates inventory, generates an invoice, and records the transaction in the finance department.

# ERP vs CRM

| ERP                                        | CRM                                            |
|--------------------------------------------|------------------------------------------------|
| Focuses on internal business operations    | Focuses on customer relationships              |
| Manages finance, HR, inventory, production | Manages sales, marketing, and customer service |
| Improves operational efficiency            | Improves customer satisfaction                 |
| Used by employees across departments       | Used mainly by sales and support teams         |

## Conclusion

ERP manages business operations, whereas CRM helps organizations build and maintain strong customer relationships.

# Business Strategy and IT Strategy

Business Strategy defines the long-term goals of an organization, while IT Strategy explains how technology will help achieve those goals.

## Business Strategy

- Increase revenue
- Expand market share
- Improve customer experience
- Reduce operational costs

## IT Strategy

- Implement cloud computing
- Strengthen cybersecurity
- Upgrade business software
- Improve network infrastructure

## Alignment Capability:

- Good communication between IT and business teams
- Strong leadership and collaboration
- Shared goals and understanding
- Flexible processes
- Governance and change management

## Importance

Aligning business and IT strategies helps organizations achieve better performance and sustainable growth.

# How They Work Together

## **A. Business Strategy → Drives IT Strategy**

- Business goals determine what IT systems are needed.

Example:

If the business strategy focuses on customer experience → IT invests in CRM, chatbots, website optimization.

## **B. IT Strategy → Enables Business Strategy**

- IT provides the tools and infrastructure to execute the business plan.

Example:

If the business wants global expansion → IT sets up cloud platforms, e-commerce systems, and ERP modules.

## **C. Alignment Capability → Ensures Continuous Fit**

- Business environment and technology both change.

Alignment capability ensures the two strategies stay coordinated over time through:

- Regular reviews
- Cross-department communication
- Agile adjustments
- Shared decision-making



# Types of Data

Organizations work with different types of data depending on how it is organized.

| Type                 | Description                    | Example                |
|----------------------|--------------------------------|------------------------|
| Structured Data      | Organized in rows and columns  | Student database       |
| Semi-Structured Data | Partially organized using tags | XML, JSON              |
| Unstructured Data    | No fixed format                | Images, videos, emails |

## Why It Matters

Understanding data types helps organizations choose the right storage and analysis methods.

# Data Collection

**Data collection** is the process of **gathering information** from various sources to **answer questions, test hypotheses, or make decisions**. It is used in **research, business, education, government, and data analytics**.

## Types of Data (Based on Nature)

### a) Qualitative Data (Non-numerical)

- Descriptive, textual information
- Answers “**why**” and “**how**”

Examples: opinions, feedback, comments, interview responses

### b) Quantitative Data (Numerical)

- Data in numbers
- Answers “how much”, “how many”

Examples: marks, sales figures, age, income, rating (1–5)

Data collection methods are often chosen based on whether you need **qualitative** or **quantitative** data.

# Data Collection Methods

Data collection is the process of gathering information for analysis and decision-making.

## **Primary Data**

Collected directly from the original source.

### **Examples**

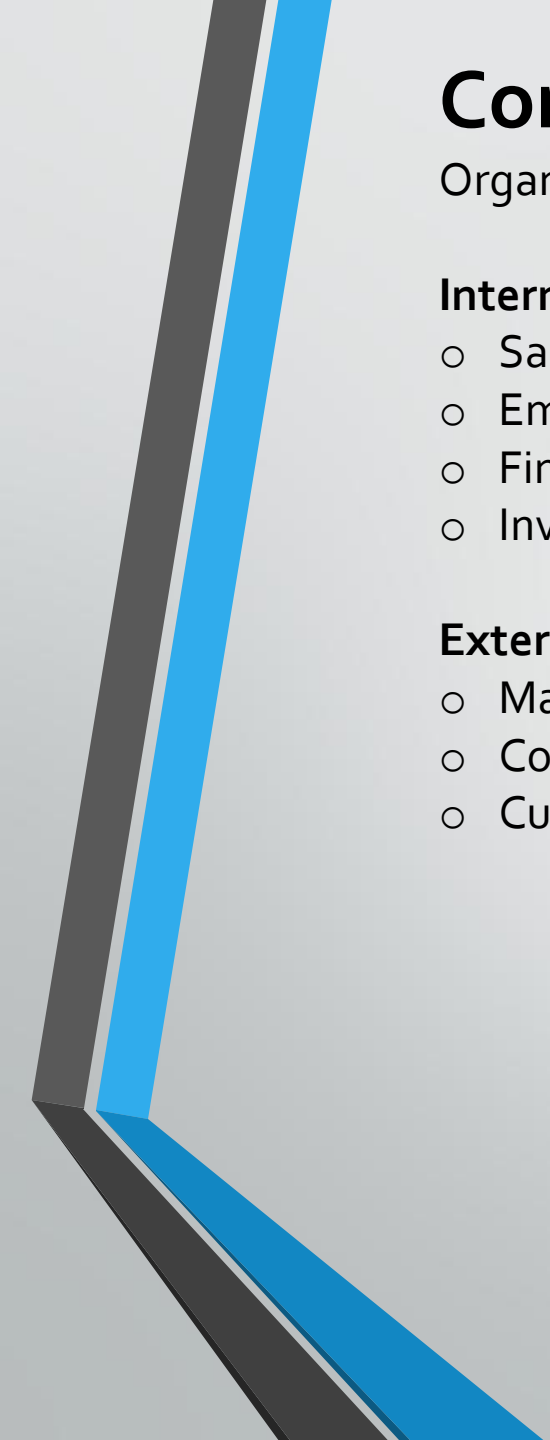
- Surveys
- Interviews
- Questionnaires
- Observations

## **Secondary Data**

Collected from existing sources.

### **Examples**

- Books
- Government reports
- Research papers
- Company databases



# Common Data Sources

Organizations collect data from various sources to improve business decisions.

## Internal Data

- Sales reports
- Employee records
- Financial statements
- Inventory data

## External Data

- Market trends
- Competitor information
- Customer demographics

## Open Data

- Government statistics
- Weather reports
- Census data

## Third-Party Data

- Google Analytics
- Marketing reports
- Industry research

# Data Life Cycle

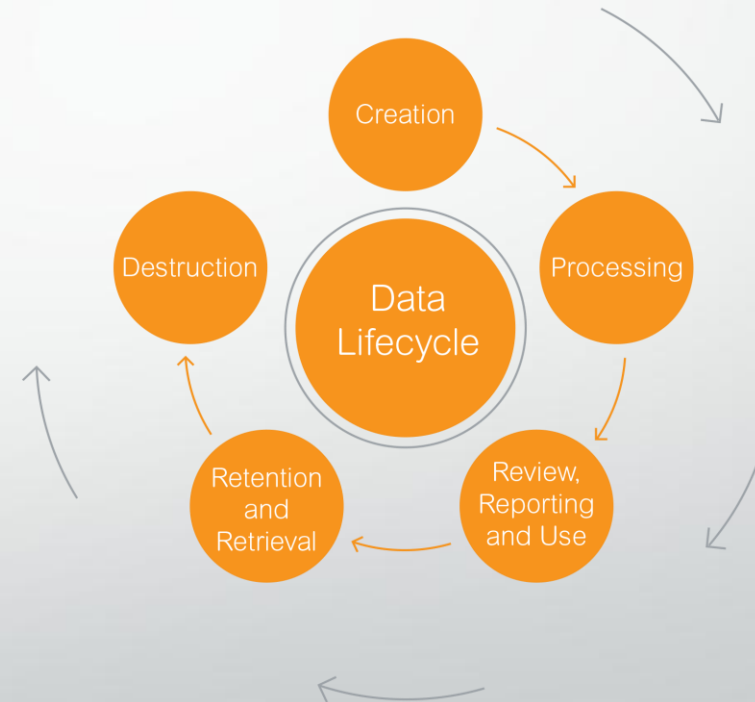
The Data Life Cycle describes how data is managed from creation to deletion.

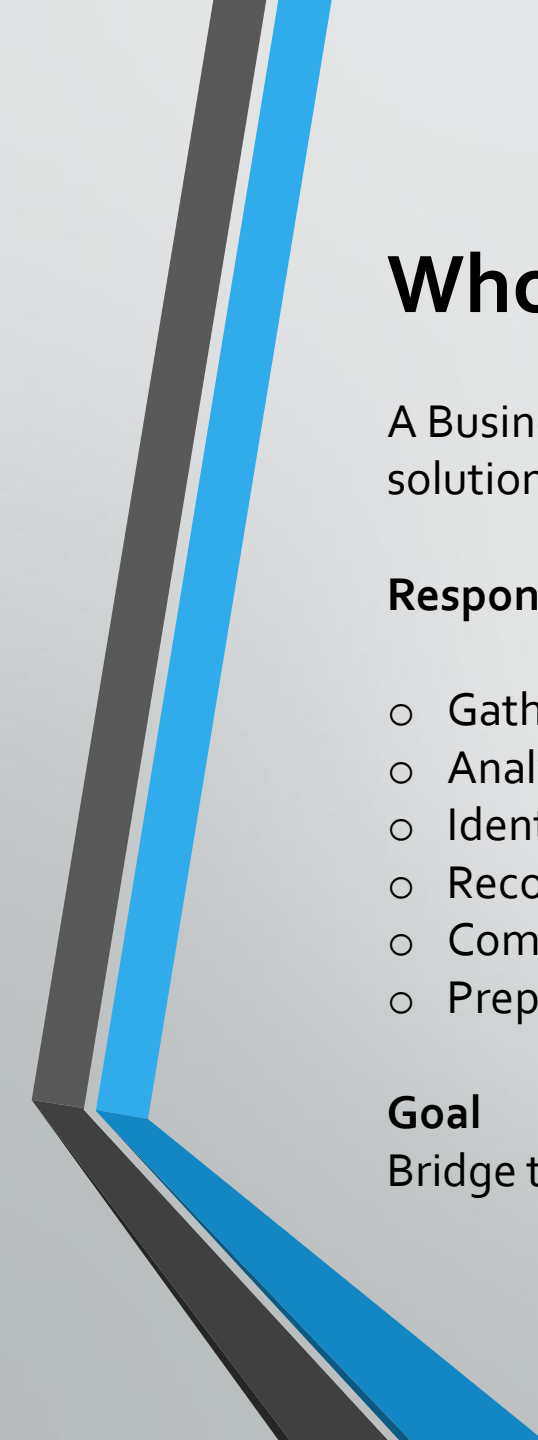
## Stages

- i. Data Creation
- ii. Data Processing
- iii. Data Review, Reporting and Use
- iv. Data Retention and Retrieval
- v. Data Deletion / Destruction

## Importance

- Better data management
- Improved security
- Regulatory compliance
- Efficient storage





# Who is a Business Analyst?

A Business Analyst (BA) helps organizations understand business needs and identify technology-based solutions to improve performance.

## Responsibilities

- Gather business requirements
- Analyse data
- Identify problems
- Recommend solutions
- Communicate with stakeholders
- Prepare reports

## Goal

Bridge the gap between business objectives and technology solutions.

# Key responsibilities:

- **Analyzing business needs:** They research and identify the organization's functional and technical needs and requirements to solve problems or capitalize on opportunities.
- **Data analysis:** They analyze large datasets using tools like SQL and Excel to find trends, create reports, and build financial models to support decision-making.
- **Process improvement:** They evaluate existing processes, identify inefficiencies, and propose solutions, which can include recommending new technology, policies, or system changes.
- **Requirements gathering:** They collect detailed requirements from stakeholders and use them to define the scope of projects.
- **Communication and documentation:** They create reports and presentations to communicate their findings to senior management and other stakeholders.
- **Solution implementation:** They may help oversee the implementation of new systems or processes and ensure they meet the defined requirements and provide value to the business.



# Skills of a Business Analyst

A successful Business Analyst requires both technical and interpersonal skills.

## Technical Skills

- SQL
- Microsoft Excel
- Data Analysis
- Process Modelling
- Documentation

## Soft Skills

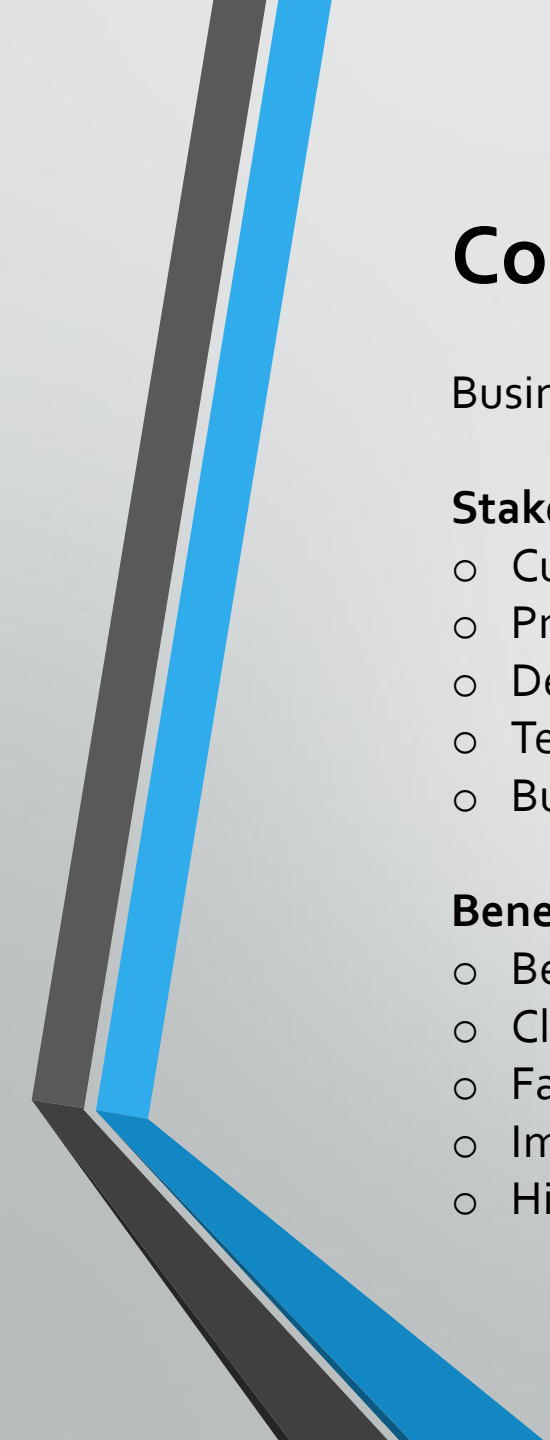
- Communication
- Problem-solving
- Critical thinking
- Teamwork
- Time management



# Tools Used by Business Analysts

Business Analysts use different tools to collect, analyse, and present information.

| Tool               | Purpose                |
|--------------------|------------------------|
| Microsoft Excel    | Data analysis          |
| JIRA               | Project management     |
| Confluence         | Documentation          |
| Tableau            | Data visualization     |
| Microsoft Visio    | Process diagrams       |
| Lucidchart         | Flowcharts             |
| Microsoft Power BI | Interactive dashboards |



# Collaboration in Business Analysis

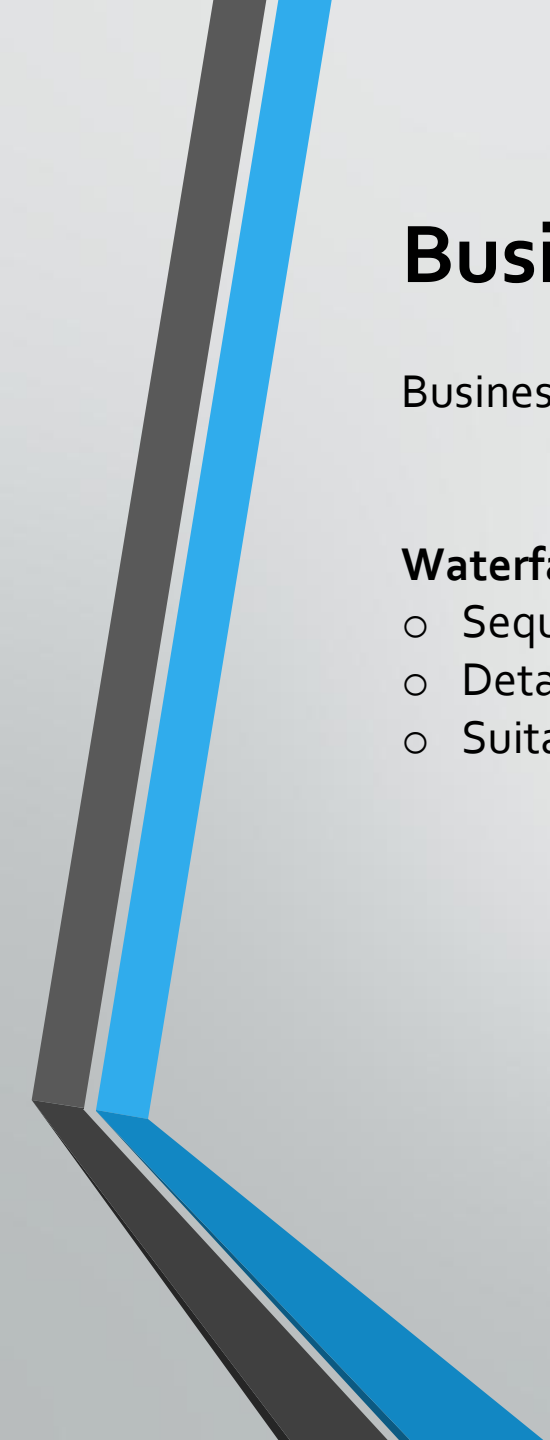
Business Analysts work closely with different stakeholders throughout a project.

## **Stakeholders**

- Customers
- Project Managers
- Developers
- Testers
- Business Owners

## **Benefits of Collaboration**

- Better communication
- Clear project requirements
- Faster problem-solving
- Improved project success
- Higher customer satisfaction



# Business Analysis Methodologies

Business Analysis follows structured approaches to manage projects effectively.

## Waterfall

- Sequential process
- Detailed planning
- Suitable for fixed requirements

## Agile

- Flexible approach
- Continuous feedback
- Frequent software releases
- Ideal for changing requirements

# Waterfall Methodology

## What is Waterfall?

Waterfall is a traditional project management approach where each phase is completed before moving to the next. It follows a structured and sequential process.

## Phases

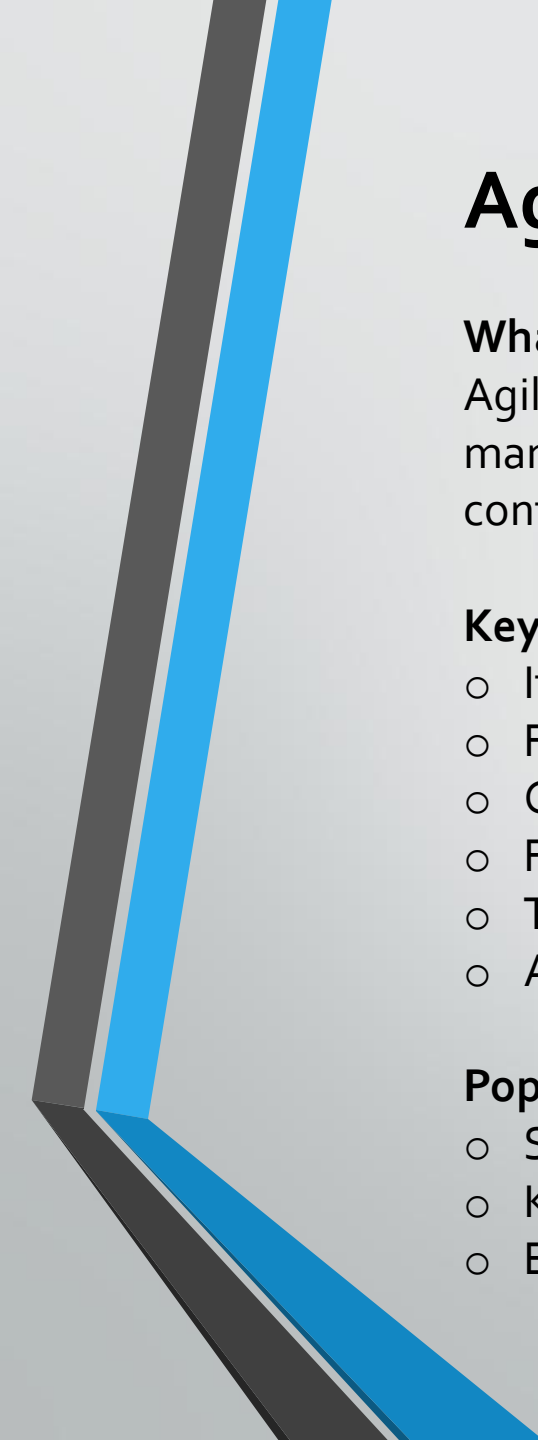
- Gather **complete requirements** at the beginning.
- Prepare **detailed documentation** (BRD, SRS, use cases).
- Get sign-off from stakeholders **before development starts**.
- Limited scope for change later.

## Advantages

- Clear structure and planning.
- Good for **stable, well-understood requirements**.
- Strong documentation.

## Disadvantages

- Not flexible to changes.
- Feedback comes **late**, after development.
- Risk of building something that doesn't match user needs if requirements change.



# Agile Methodology

## What is Agile?

Agile is a flexible project management methodology that divides a project into small, manageable iterations called **sprints**. It emphasizes teamwork, customer feedback, and continuous improvement.

## Key Features

- Iterative development
- Frequent customer feedback
- Continuous testing
- Faster delivery
- Team collaboration
- Adaptable to changing requirements

## Popular Agile Frameworks

- Scrum
- Kanban
- Extreme Programming (XP)

# Agile vs Waterfall

| Basis          | Waterfall Model                                                          | Agile Model                                           |
|----------------|--------------------------------------------------------------------------|-------------------------------------------------------|
| Approach       | Linear and sequential                                                    | Iterative and incremental                             |
| Project Phases | Fixed phases: Requirements → Design → Development → Testing → Deployment | Work done in small cycles called sprints              |
| Requirements   | Defined fully at the beginning; changes are difficult later              | Requirements can change anytime; refined continuously |
| Flexibility    | Low – not suitable for frequent changes                                  | High – designed to handle changing needs              |

# Agile vs Waterfall

|                             |                                                        |                                                            |
|-----------------------------|--------------------------------------------------------|------------------------------------------------------------|
| <b>Customer Involvement</b> | Mainly at start (requirements) and end (delivery)      | Continuous involvement in every sprint (reviews, feedback) |
| <b>Delivery</b>             | Final product delivered at the end of the project      | Working software delivered frequently (after each sprint)  |
| <b>Documentation</b>        | Heavy, detailed documentation (BRD, SRS, etc.)         | "Just enough" documentation; more focus on communication   |
| <b>Testing</b>              | Testing happens after development is completed         | Testing happens within each sprint (early and continuous)  |
| <b>Risk Handling</b>        | Risks discovered late (towards testing/implementation) | Risks identified and handled early in each iteration       |
| <b>Best Suited For</b>      | Projects with clear, stable requirements               | Projects with uncertain or changing requirements           |

# Stakeholder Management

## What is Stakeholder Management?

Stakeholder management is the process of identifying, communicating with, and managing individuals or groups affected by a project.

## Common Stakeholders

- Customers
- Employees
- Project Managers
- Investors
- Suppliers
- Government Agencies

## Benefits

- Better communication
- Reduced project risks
- Improved collaboration
- Higher project success rate



# Scenario Analysis

## What is Scenario Analysis?

Scenario analysis is a planning technique used to evaluate different future situations and their possible impact on business decisions.

## Types of Scenarios

- Best Case
- Most Likely Case
- Worst Case

## Applications

- Business planning
- Financial forecasting
- Risk assessment
- Investment decisions

## Example

A company estimates profits under different sales conditions before launching a new product.

# Data Governance

## What is Data Governance?

Data governance is a framework of policies and procedures that ensures data is managed securely, accurately, and consistently across an organization.

## Objectives

- Improve data quality
- Protect sensitive information
- Define user responsibilities
- Ensure legal compliance
- Maintain data consistency

## Benefits

- Reliable information
- Better decision-making
- Enhanced security
- Reduced data errors

# Ethical Use of Data

## What is Ethical Data Use?

Ethical data use means collecting, storing, and using information responsibly while respecting individuals' privacy and rights.

## Principles

- Obtain user consent
- Maintain transparency
- Protect privacy
- Avoid discrimination
- Ensure accountability
- Use data only for intended purposes

## Importance

Ethical practices build trust between organizations and users.

# GDPR vs CCPA

Both the General Data Protection Regulation (GDPR) and the California Consumer Privacy Act (CCPA) aim to give individuals greater control and transparency over their personal data, but they differ significantly in scope, core principles, and enforcement. Both aim to protect individuals' personal data and give them control over how businesses use it.

| GDPR                                   | CCPA                                           |
|----------------------------------------|------------------------------------------------|
| European Union regulation              | California, USA regulation                     |
| Protects personal data                 | Protects consumer privacy                      |
| Requires user consent                  | Allows users to opt out of data selling        |
| Applies to many organizations globally | Applies to qualifying businesses in California |

# Key Similarities Between GDPR and CCPA

**Goal:** Both regulations share the primary objective of protecting individual data privacy rights and enhancing transparency in data handling practices.

**Global Reach:** Both laws can apply to organizations worldwide, provided they process the personal data of EU residents (GDPR) or California residents (CCPA), regardless of the business's physical location.

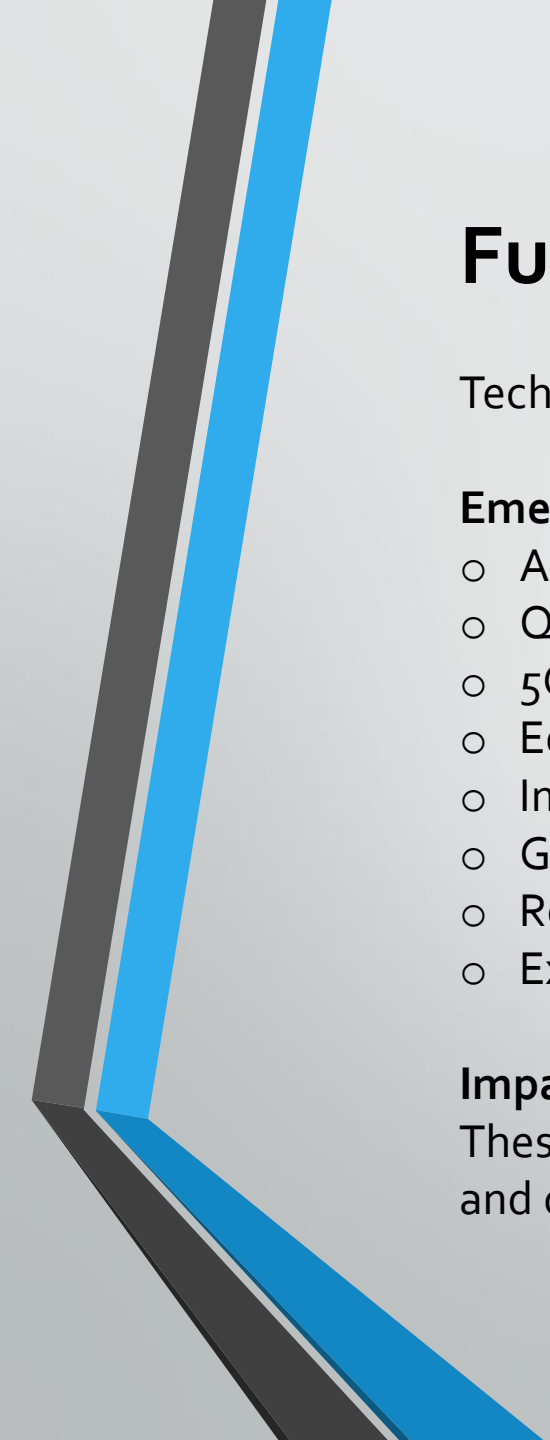
**Consumer Rights:** Both grant individuals similar core rights concerning their data, including the right to know what data is collected, the right to access that data, and the right to request its deletion (right to erasure).

**Security Measures:** Both require organizations to implement reasonable security measures to protect personal data and mandate breach notifications to affected individuals.

**Penalties:** Both laws impose significant fines for non-compliance, aiming to deter violations and ensure accountability.

# Key Differences Between GDPR and CCPA

| Feature           | GDPR (General Data Protection Regulation)                                | CCPA (California Consumer Privacy Act)                                                                    |
|-------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Region            | European Union (EU)                                                      | California, United States                                                                                 |
| Purpose           | Protects personal data and privacy                                       | Protects consumer privacy and data rights                                                                 |
| Who is Protected? | Individuals in the EU                                                    | California residents                                                                                      |
| Consent           | Requires user consent before collecting or processing personal data      | Allows data collection but gives consumers the right to opt out of the sale of their personal information |
| User Rights       | Access, correct, delete, restrict processing, and transfer personal data | Access, delete, know what data is collected, and opt out of data selling                                  |
| Business Scope    | Applies to organizations processing EU residents' data                   | Applies to qualifying businesses operating in California                                                  |



# Future Trends in Information Technology

Technology continues to evolve rapidly, creating new opportunities and challenges.

## Emerging Trends

- Artificial Intelligence
- Quantum Computing
- 5G Networks
- Edge Computing
- Internet of Things
- Green Computing
- Robotics
- Extended Reality (AR/VR)

## Impact

These technologies will transform healthcare, education, manufacturing, transportation, and communication.

# Challenges in Information Technology

Despite its advantages, IT also faces several challenges.

## Major Challenges

- Cybersecurity threats
- Data privacy concerns
- Rapid technological changes
- High implementation costs
- Skill shortages
- System failures
- Digital divide
- Environmental impact of e-waste

## Solutions

- Continuous learning
- Strong security measures
- Regular system updates
- Sustainable IT practices



# Virtualization Technology

## What is Virtualization?

Virtualization is a technology that allows multiple virtual computers or operating systems to run on a single physical machine. It improves hardware utilization and reduces infrastructure costs.

## Types of Virtualization

- Server Virtualization
- Desktop Virtualization
- Storage Virtualization
- Network Virtualization

## Advantages

- Better resource utilization
- Reduced hardware costs
- Easy backup and recovery
- Improved scalability
- Simplified system management

## Examples

- VMware
- Oracle VirtualBox
- Microsoft Hyper-V

# Disaster Recovery and Backup

## What is Disaster Recovery?

Disaster Recovery (DR) is the process of restoring IT systems and data after unexpected events such as cyberattacks, hardware failures, or natural disasters.

## Common Backup Methods

- Full Backup
- Incremental Backup
- Differential Backup
- Cloud Backup

## Importance

- Prevents data loss
- Ensures business continuity
- Reduces downtime
- Protects critical information
- Speeds up system recovery

# Careers in Information Technology

The IT industry offers a wide range of career opportunities for professionals with different skills and interests.

## Popular Career Options

- Software Developer
- Web Developer
- Network Engineer
- Database Administrator (DBA)
- Cybersecurity Analyst
- Cloud Engineer
- Data Analyst
- AI & Machine Learning Engineer
- DevOps Engineer
- System Administrator

## Essential Skills

- Programming (Java, Python, C++)
- Networking
- Database Management
- Cloud Computing
- Cybersecurity
- Communication & Problem-solving

## Future Scope

The demand for IT professionals continues to grow as organizations adopt digital technologies, cloud services, artificial intelligence, and data-driven solutions.



**THANK YOU**